



First measurement of the electric formfactor of the neutron in the exclusive quasielastic scattering of polarized electrons from polarized ^3He

M. Meyerhoff^{a,1}, D. Eyl^{b,1}, A. Frey^{b,1}, H.G. Andresen^b, J.R.M. Annand^c, K. Aulenbacher^a, J. Becker^a, J. Blume-Werry^b, Th. Dombó^b, P. Drescher^a, J.E. Ducret^b, H. Fischer^a, P. Grabmayr^d, S. Hall^c, P. Hartmann^b, T. Hehl^d, W. Heil^a, J. Hoffmann^a, J.D. Kellie^c, F. Klein^b, M. Leduc^e, H. Möller^b, Ch. Nachtigall^a, M. Ostrick^b, E.W. Otten^a, R.O. Owens^c, S. Plützer^a, E. Reichert^a, D. Rohe^a, M. Schäfer^a, L.D. Schearer^{f,2}, H. Schmieden^b, K.-H. Steffens^b, R. Surkau^a, Th. Walcher^b

^a Institut für Physik, University of Mainz, Mainz, Germany

^b Institut für Kernphysik, University of Mainz, Mainz, Germany

^c Department of Physics and Astronomy, University of Glasgow, Glasgow, Scotland

^d Physikalisches Institut, University of Tübingen, Germany

^e École Normale Supérieure, Paris, France

^f University of Missouri, Rolla, USA

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Abstract

A first measurement of the asymmetry in quasielastic scattering of longitudinally polarized electrons from a polarized ^3He gas target in coincidence with the knocked out neutron is reported. This measurement was made feasible by the cw beam of the 855 MeV Mainz Microtron MAMI. It allows a determination of the electric formfactor of the neutron G_E^n independent of binding effects to first order. At $\bar{Q}^2 = 0.31 \text{ (GeV}/c)^2$ two asymmetries $\bar{A}_{\parallel}(S_{\text{He}} \parallel \mathbf{q})$ and $\bar{A}_{\perp}(S_{\text{He}} \perp \mathbf{q})$ have been measured giving $\bar{A}_{\parallel} = (-7.40 \pm 0.73)\%$ and $\bar{A}_{\perp} = (0.89 \pm 0.30)\%$. The ratio $\bar{A}_{\perp}/\bar{A}_{\parallel}$ is independent of the absolute value of the electron and target polarization and yields $G_E^n = 0.035 \pm 0.012 \pm 0.005$.

The question of the electromagnetic structure of the nucleon is of fundamental importance both in nuclear and particle physics in order to make significant statements on quark wave functions or models of the nucleon. While the data on the proton have over many

years gradually been extended in momentum transfer and precision, comparatively little progress has been made on the neutron charge and magnetic formfactors. In particular the experimental information on the charge formfactor G_E^n is still rather limited. Our present knowledge on G_E^n is based on three experimental sources:

(i) In thermal neutron scattering from atomic elec-

¹ comprises parts of doctoral theses

² deceased 7.3.1993

trons [1,2] a mean squared charge radius of $\langle r_n^2 \rangle \simeq -0.116 \text{ fm}^2$ is inferred giving a value of the slope $dG_E^n/dQ^2 \simeq 0.019 \text{ fm}^2$ at $Q^2 \approx 0$. It clearly shows that G_E^n is finite and represents a constraint to measurements at larger Q^2 , needed for a determination of the distribution of charge in space.

(ii) In the medium momentum transfer range ($0.2 \leq Q^2 \leq 0.6 \text{ (GeV/c)}^2$) the longitudinal structure function $A(Q^2)$ in elastic electron deuteron scattering has been determined [3,4] containing also G_E^n among other dominating contributions. Though this method is capable of delivering good statistics data it is hampered by a model dependence of up to a factor of 2.

(iii) At high momentum transfer ($1.75 \leq Q^2 \leq 4 \text{ (GeV/c)}^2$) a Rosenbluth separation of the quasielastic electron-deuteron scattering is possible [5] which does not have the model dependence of the method of (ii).

An improved experimental method of determining G_E^n is required and can now be provided in polarized electron-neutron ($\vec{e}-\vec{n}$) scattering. In contrast to the unpolarized cross section measurement, for which the effect of the electric scattering cross section is submerged by the much larger magnetic one, the polarization asymmetry has a component being proportional to an interference term between the electric and magnetic scattering amplitudes. More precisely, the asymmetry of the cross section for $e-n$ scattering with respect to the electron helicity P_e responds differently to the transverse polarization component of the neutron $p_x = P_n \cdot \sin(\theta^*) \cdot \cos(\phi^*)$ and to the longitudinal one $p_z = P_n \cdot \cos(\theta^*)$. θ^* and ϕ^* are the Euler angles of the neutron polarization P_n , i.e. θ^* with respect to the momentum transfer q and ϕ^* with respect to the scattering plane. The asymmetry A is then given by [6]

$$\begin{aligned}
 A = & -P_e \left[2\sqrt{\tau(1+\tau)} \tan\left(\frac{\vartheta}{2}\right) G_E^n G_M^n p_x + \right. \\
 & \left. 2\tau \sqrt{1+\tau + (1+\tau)^2 \tan^2\left(\frac{\vartheta}{2}\right)} \tan\left(\frac{\vartheta}{2}\right) (G_M^n)^2 p_z \right] \\
 & \times \left[(G_E^n)^2 + (G_M^n)^2 \cdot (\tau + 2\tau(1+\tau) \cdot \tan^2\left(\frac{\vartheta}{2}\right)) \right]^{-1} \\
 = & A_\perp \sin \theta^* \cos \phi^* + A_\parallel \cos \theta^* \quad (1)
 \end{aligned}$$

with $\tau = Q^2/4 \cdot M^2$ and ϑ being the scattering angle of the electron. One recognizes that only $A_\perp$ is sensitive to the interference of G_E^n and G_M^n .

Polarized studies with ^3He have received much attention because the ^3He contains an approximately saturated spin pair of two protons and, therefore, the quasielastic scattering from the neutron is dominated by the magnetic part of the interaction. This simple argument is corroborated by more detailed theoretical considerations, that in the quasielastic scattering region the spin-dependent properties are dominated by the neutron within the nucleus [7]. Thus the elastic formfactors of the neutron can be studied through inclusive $^3\text{He}(\vec{e}, e')$ or exclusive $^3\text{He}(\vec{e}, e'n)$ reactions. Inclusive experiments have already been carried out at Bates [8,9], but the evaluation of the proton contribution to the asymmetry is model dependent and a reliable extraction of the neutron charge properties is difficult [10]. The cw, high current polarized beam of the accelerator MAMI at Mainz offers the possibility for exclusive $^3\text{He}(\vec{e}, e'n)$ and also $D(\vec{e}, e'\vec{n})$, which is an alternative method of measuring G_E^n . Both experiments which are performed with a similar detector system [11], aim at a precision of $\pm 10\%$ in the range of momentum transfer $Q^2 = 0.2$ to 0.6 (GeV/c)^2 . They provide complementary results which will allow a check of nuclear medium effects on the formfactors, as well as the control of residual systematic errors. Data on deuterium have also been taken. A preliminary account can be found in Ref. [12].

In this paper we report on an exploratory experiment on ^3He intended to provide a full check of the experimental technique and also to deliver first results. Polarized electrons, produced by laser induced photoemission on a GaAsP crystal are injected into the microtron MAMI [13], where they are accelerated to 855 MeV. A spin rotator [13] in the injection line allows the adjustment of the electron spin to its longitudinal direction at target position independent of spin precession. During the experiment the electron helicity was reversed statistically with a frequency of 1 Hz to eliminate instrumental asymmetries. The average beam polarization was measured to be $\bar{P}_e = (32 \pm 3)\%$ using a Møller polarimeter [15].

The set-up for the electron and neutron detection is shown schematically in Fig. 1. It represents a subset of the full detector arrays, providing around 20% of the final detector acceptance. Scattered electrons are detected in a segmented Pb-glass calorimeter which has a angular resolution of $\delta\vartheta_e = \delta\phi_e = 0.3^\circ$. Its energy resolution of $\delta E/E \simeq 16\%/\sqrt{E(\text{GeV})}$ is more or

set-up shows that Q can be reconstructed with an accuracy of $\pm 5\%$ for the $e'n$ final state taking into account the resolution of each detector. In the case of the ${}^3\text{He}(\bar{e}, e'n)pp$ reaction the kinematics is in principle incomplete. Still we can reconstruct Q , since the quasielastic cross section is dominated by the quasi two body breakup with little relative energy in the pp -system as we infer from an analogue ${}^3\text{He}(e, e'p)np$ measurement [19].

To carry out the experiment a new type of a dense polarized ${}^3\text{He}$ gas target was developed [20,21]. The ${}^3\text{He}$ gas is spin polarized by direct optical pumping in its metastable ($1s2s\ {}^3S_1$) state at pressures around 1 mb [22]. In order to reach the high densities required for this experiment, the ${}^3\text{He}$ -gas is compressed by means of a Toepler pump in which a mercury column in a glass cylinder is activated every 27s to compress the gas from the optical pumping cell (OPC) into the target cell (TC) reaching pressures around 1 bar. The whole apparatus is immersed in a homogeneous magnetic guiding field of 8.3 Gauss which sets the ${}^3\text{He}$ spin alignment perpendicular to the momentum transfer q . In the OPC a nuclear spin polarization of $P \simeq 60\%$ is achieved at a throughput of $\Gamma_p = 1.2 \times 10^{18}$ atoms/s. On the way back from the TC to the OPC the gas passes through a getter purifier. It ensures a spectroscopically clean ${}^3\text{He}$ discharge in the OPC by filtering out in particular a relative nitrogen concentration of $[N_2]/[{}^3\text{He}] \simeq 10^{-4}$ by which the target gas is vaccinated after compression. The nitrogen gas serves to quench the formation of molecular ${}^3\text{He}_2^+$ ions which is the dominant relaxation mode under charged particle beam conditions [23]. An additional set of coils placed around the TC can be activated to rotate the spin to a direction parallel to q . In this mode of operation the compression cycle has to be stopped which causes a slight decrease of the target polarization with a time constant of $T_1 = 3$ h. Fig. 2 shows the target polarization measured by NMR at a beam current of $I = 10\ \mu\text{A}$. The polarization was fairly constant during 100 h of beam time. The periodic structure is caused by the alternate settings of the target spin S_{He} parallel or perpendicular to q . On the average we measured a target polarization of $\bar{P}_{\text{He}} = 38\%$.

For this exploratory experiment the electron arm with its angular acceptance of $\Delta\vartheta_e = \pm 3.9^\circ$ and $\Delta\varphi_e = \pm 4.5^\circ$ was placed at an angle of $\vartheta_e = 43^\circ$ which gave a central momentum $|q_c| = 586\ \text{MeV}/c$ for an aver-

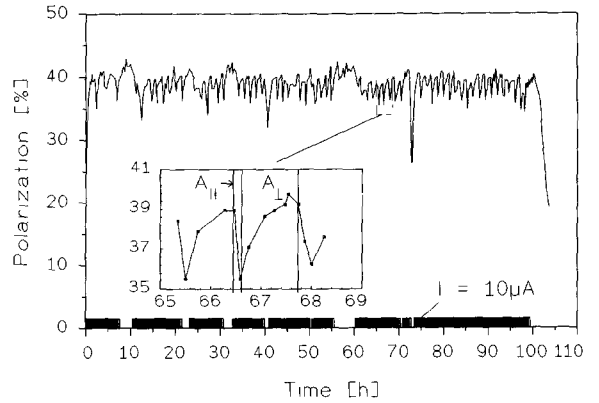


Fig. 2. Polarization in the ${}^3\text{He}$ -target cell during a run at $I = 10\ \mu\text{A}$. The periodic structure results from the alternate settings of the target spin orientation parallel or perpendicular to q with $t_{||} = 10$ min and $t_{\perp} = 60$ min respectively (see inset). The 'beam-on' and 'beam-off' periods demonstrate the influence of the electron beam on the target depolarization (2% loss absolute).

age four momentum transfer $\bar{Q}^2 = 0.31\ (\text{GeV}/c)^2$. The neutron detector centered at $\bar{\vartheta}_n = 53^\circ$ covered a solid angle of $\Delta\Omega = 62\ \text{msr}$ ($\Delta\vartheta_n = \pm 4.1^\circ$, $\Delta\varphi_n = \pm 12.1^\circ$) and had a detection efficiency of $\bar{\epsilon}_n \simeq 25\%$ for neutrons in the energy range of $E_n \approx 150\ \text{MeV}$. Hence the total detection efficiency for neutrons associated with a quasielastic electron event in the lead glass calorimeter is $\bar{\epsilon}_{\text{tot}} = \bar{\epsilon}_n \times (1 - \delta_{\text{loss}}) \times \bar{\epsilon}_{\text{geo}} \simeq 7\%$, where $\bar{\epsilon}_{\text{geo}} = 0.45$ is the calculated acceptance of the neutron detector taking into account the Fermi momentum of the bound neutron [19]. For the kinematics used, an effective target length of $l_{\text{eff}} = 10\ \text{cm}$ could be imaged onto the PM of the air Čerenkov detector resulting in a thickness of $2.4 \cdot 10^{20}\ \text{cm}^{-2}$ at $p = 940\ \text{mb}$. With the coincidence condition $(\text{Pb}_{\text{glass}} \wedge \check{\text{C}}_{\text{air}}) \wedge N_{\text{det}} \wedge \overline{\text{Veto}}$ data were written on tape with a trigger rate of $\Gamma_{\text{trigger}} \simeq 2.5\ \text{Hz}$ at $I = 10\ \mu\text{A}$. Altogether a total of $1027\ \mu\text{A} \cdot \text{h}$ of charge was collected. Using the elementary $e-n$ cross sections the quasielastic event rate for the exclusive $(e, e'n)$ reaction is expected to be $\sim 0.5\ \text{Hz}$. In the off-line analysis, the pulse height from the electron and neutron detectors, the TOF in the neutron detector and kinematical correlations were used to filter out the inelastic and accidental events. In Fig. 3 the resultant pulse height spectrum (1) of scattered electrons in the lead glass calorimeter shows a dominant quasielastic peak with a low energy shoulder due to inelastic events. Evidently a complete sep-

Table 1

Results of asymmetry measurements at $\bar{Q}^2 = 0.31 \text{ (GeV/c)}^2$ in exclusive ${}^3\text{He}(\bar{e}, e'n)$

Orientation	Charge ($\mu\text{A} \cdot \text{h}$)	θ_c^* (deg)	ϕ_c^* (deg)	Asymmetry (%)
A	451	88.2	-1.8	$+0.44 \pm 0.42$
B	431	88.2	181.9	-1.35 ± 0.44
	882 (combined)			$(A_A - A_B)/2 = \bar{A}_\perp = +0.89 \pm 0.30$
C	77	2.1	-56.2	-7.22 ± 1.01
D	68	177.7	-56.2	$+7.62 \pm 1.07$
	145 (combined)			$(A_C - A_D)/2 = \bar{A}_\parallel = -7.40 \pm 0.73$
				ratio: $\bar{A}_\perp/\bar{A}_\parallel = -0.120 \pm 0.042 \pm 0.01$

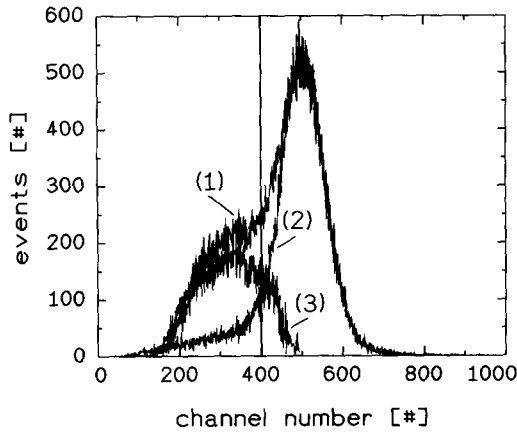


Fig. 3. Pulse height spectra of scattered electrons in the lead glass calorimeter measured in exclusive ${}^3\text{He}(\bar{e}, e'n)$ (1) and $H_2(e, e'p)$ (2) under the same kinematical conditions. The difference spectrum (3) is used as guideline for an appropriate pulse height cut, which is set at channel 400 (vertical line).

aration between both types of events is not given. For comparison an equivalent spectrum (2) obtained in a $H_2(e, e'p)$ calibration run, made at identical kinematics, is included. The observed peak width is mainly determined by the lead glass energy resolution which is 21%. The difference spectrum (curve (3) in Fig. 3) proves that inelastic contributions to the quasielastic signal will not exceed 8% above an appropriate threshold channel of 400. For the analysis of the exploratory experiment we therefore content ourselves with the limited energy resolution of the lead glass calorimeter; an energetically better selection of the quasifree $e-n$ scattering by reconstructing Q was not made. The resultant quasielastic event rate of $\Gamma = 0.4 \text{ Hz}$ at $I = 10 \mu\text{A}$ is in accordance with the one calculated above.

The asymmetry was measured for four different

configurations of the target spin direction and the results are given in Table 1. Combining all the data (after reversing the sign of orientation B and D results) yields the asymmetries $\bar{A}_\parallel = (-7.40 \pm 0.73)\%$ and $\bar{A}_\perp = (+0.89 \pm 0.30)\%$. Deviations of the target spin orientations from directions along and normal to q_c were caused by the earth's magnetic field which was not compensated for. Its effect on the measured asymmetries in orientation A and B is evident and manifests in a common offset stemming from admixtures of the dominant $A_\parallel$ -term ($\sim (G_M^n)^2 \cdot \cos(\theta^*)$) in Eq. (1). This is cancelled, however, in the difference of the measured pairs of asymmetries $\bar{A}_\perp = (A_A - A_B)/2$. For the orientations C and D a corresponding asymmetry offset is caused by the weaker $A_\perp$ -term ($\sim G_E^n G_M^n \sin(\theta^*) \cos(\phi^*)$), but is actually too small to be seen. Anyway it is eliminated by taking the difference $\bar{A}_\parallel = (A_C - A_D)/2$.

Sources of systematic errors on the data may be split into 4 categories:

(a) those that arise from the uncertainty in the absolute calibration of the beam polarization ($\pm 10\%$) and target polarization ($\pm 8\%$),

(b) those that arise from a dilution of the quasielastic signal from some background sources (i.e. empty target contribution $\simeq 5\%$),

(c) those that involve the spindependent asymmetry directly, namely systematic uncertainties due to inelastic contributions to the quasielastic signal. We looked for asymmetries in the lower part (channel: < 400) of the pulse height spectrum giving $\bar{A}_\perp(< 400) = (0.95 \pm 0.47)\%$ and $\bar{A}_\parallel(< 400) = (-1.25 \pm 1.12)\%$ respectively. Since inelastic events only contribute $\simeq 8\%$ to our quasielastic signal, we allow a systematic error of $\pm 10\%$ on the asymmetry ratio $\bar{A}_\perp/\bar{A}_\parallel$ (see

Table 1) and

(d) contributions of conversion neutrons stemming from (p, n) charge exchange reactions in the lead wall which have been investigated in a separate test run using a LH_2 -target [24]. From these measurements we conclude that their relative contributions are less than 3%, thus being negligibly small in comparison.

With our choice to measure the longitudinal and normal component of the spin observables quasi-simultaneously (see Fig. 2) many of these systematic errors cancel.

In PWIA and quasifree kinematics $\bar{A}_{\parallel}$ is given in the limit of a free neutron (see Eq. (1)) by

$$\bar{A}_{\parallel} = P_e \cdot P_n \cdot f_{\parallel} \cdot (1 + \mathcal{O}((G_E/G_M)^2)) \quad (2)$$

with

$$f_{\parallel} = -\frac{2\sqrt{1 + \tau + (1 + \tau)^2 \tan^2(\frac{\vartheta}{2})} \cdot \tan(\frac{\vartheta}{2})}{1 + 2 \cdot (1 + \tau) \cdot \tan^2(\frac{\vartheta}{2})}$$

To first order this asymmetry only depends on kinematical factors in $f_{\parallel}$ with no nucleon formfactor involved and serves to calibrate the experiment.

For the extraction of G_E^n it is advantageous to take the ratio $\bar{A}_{\perp}/\bar{A}_{\parallel}$ as given in Table 1. In this way the dominant sources of systematic errors, i.e. category (a) and (b) drop out. In order to derive a value on G_E^n from this experiment we rely on the following arguments: (i) Final state interactions and meson exchange currents will contribute additively to the leading terms in the asymmetry. Theoretical calculations show, however, that these contributions are unimportant in the quasifree reaction ${}^3\text{He}(\vec{e}, e'n)$ for $Q^2 \geq 0.3 (\text{GeV}/c)^2$. Moreover it turns out that for missing momentum components up to $p_m = 100 \text{ MeV}/c$ the calculated asymmetries do not deviate significantly from the one in pure collinear kinematic ($p_m = 0$: free neutron case) [25,26]³.

(ii) Nuclear structure effects reduce the effective neutron polarization to $P_n^{\text{eff}} \simeq 0.87 \cdot P_{\text{He}}$ [27]. Since P_n is entering the asymmetry as a factor this theoretical correction drops out in the ratio.

³ Using the measured nucleon momentum distribution from the ${}^3\text{He}(e, e'p)np$ reaction [19], a simple Fermi-gas-model calculation showed, that within the acceptance of the detector 85% of our quasielastic $(e, e'n)$ events have initial neutron momentum components of less than $100 \text{ MeV}/c$ with the maximum of the distribution at $\sim 50 \text{ MeV}/c$.

(iii) Due to the finite acceptance of the detector we have to average over the covered kinematical range. Since the neutron detector is symmetric about the q -axis, however, the integration over the azimuthal angle ϕ averages all contributions proportional to odd powers of $\cos(\phi)$ and $\sin(\phi)$ to zero. A similar argument applies to the spread $(\Delta\theta^*, \Delta\phi^*)$ about their central values θ_c^*, ϕ_c^* .

Since the asymmetries weighted by the cross sections depend linearly on the kinematic variables in the interval of acceptance, the center of gravity is preserved in averaging and the kinematic assignments of Table 1 are justified.

With these approximations we identify the measured ratio $\bar{A}_{\perp}/\bar{A}_{\parallel}$ with that of a free neutron, for which Eq. (1) yields the relation

$$G_E^n = \sqrt{\tau + \tau(1 + \tau) \tan^2(\frac{\vartheta}{2})} \cdot \frac{\bar{A}_{\perp}}{\bar{A}_{\parallel}} \cdot G_M^n \quad (3)$$

For the magnetic formfactor of the neutron the empirical dipole fit $G_M^n = \mu_N \cdot (1 + Q^2/0.71)^{-2}$ is used. This form gives G_M^n with a $\pm 10\%$ systematic accuracy at $Q^2 \simeq 0.3 (\text{GeV}/c)^2$ [28]. A recent measurement at a somewhat lower $Q^2 = 0.255 (\text{GeV}/c)^2$ confirms the value of G_M^n with this precision [29]. This systematic error has to be added to the above mentioned uncertainties. From this a total systematic error of $\pm 14\%$ follows if added quadratically. Using the values of Table 1 we obtain the final result

$$G_E^n(\bar{Q}^2 = 0.31 (\text{GeV}/c)^2) = 0.035 \pm 0.012 \pm 0.005 \quad (4)$$

As a consistency check the longitudinal asymmetry $\bar{A}_{\parallel}$ has been calculated according to Eq. (2) resulting in $\bar{A}_{\parallel}^{\text{cal}} = (-8.08 \pm 1.04)\%$, which agrees with the measured value of $\bar{A}_{\parallel}$ given in Table 1 and thus supports the approximations made for the extraction of G_E^n .

Our value of G_E^n at $\bar{Q}^2 = 0.31 (\text{GeV}/c)^2$ ($\simeq 8.1 \text{ fm}^{-2}$) is compared to the result of Platchkov et al. [4] in Fig. 4. In their fit of the data points (solid curve) using the parametrization

$$G_E^n(Q^2) = \frac{-a\mu_N\tau}{1 + b\tau} \cdot \frac{1}{(1 + Q^2/18.23[\text{fm}^{-2}])^2} \quad (5)$$

they obtained $a = 1.25 \pm 0.13$ and $b = 18.3 \pm 3.4$ for the choice of the Paris potential. The corresponding

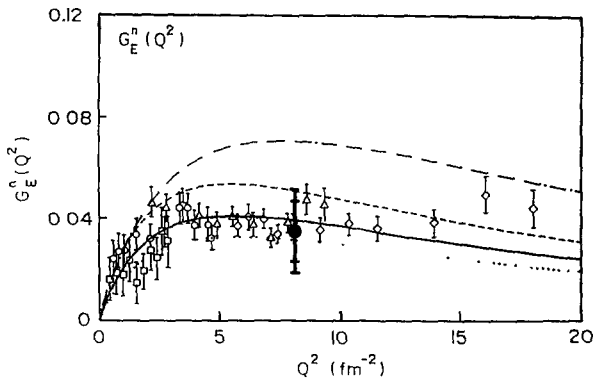


Fig. 4 Comparison of the result (full circle) of this paper with existing data on G_E^n from Ref. [4]. The solid curve gives the two parameter best fit (Eq. (5)) to the plotted data points for G_E^n as deduced using the Paris potential. In addition the corresponding two parameter fits using the RSC (dotted), Argonne V14 (dashed) or Nijmegen (dash-dotted) potentials are depicted.

curves for G_E^n using the RSC (dotted), Argonne V14 (dashed) and Nijmegen (dash-dotted) potentials are included in Fig. 4. For clarity the corresponding sets of data points are not shown. In spite of the still modest statistical accuracy of our experimental result, the Paris and RSC potential seem to be favoured and the analysis of Ref. [4] corroborated. For these two potentials the slopes of the corresponding fit curves (solid and dotted respectively) were constrained at $Q^2 = 0$ to the value derived from thermal neutron-electron scattering [2].

In summary a first measurement of G_E^n in the exclusive $^3\text{He}(\vec{e}, e'n)$ reaction at $\bar{Q}^2 = 0.31 \text{ (GeV}/c)^2$ has been performed. The result demonstrates the powerful new method of using spin observables for the investigation of the electromagnetic structure of the nucleon. With the completion of the full detector set-up a measurement of G_E^n to $\pm 10\%$ accuracy can now be envisaged in the accessible momentum transfer range of MAMI.

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